



QUEUE:

How the people are taking place?

Ans: at the backend
/ last position.
This also called rear.

Who has entered in the toilet?

Ans: First person /
the person at the
front.

**First in First Out
(FIFO)**

**Two pointer: REAR,
FRONT**

A scenario of waiting in front of a public toilet.



CSE 203: Data Structure

Data
Structure

QUEUE: First In First Out (FIFO)

EXAMPLE

Waiting to get into bus.



Waiting in a ticket counter.



Waiting in cash counter.



Waiting for voting.



Waiting for doctor.



Now say,
Who will be first to meet
the doctor?

Solution: This is maintained
by a list. Called QUEUE.



CSE 203: Data Structure

Data
Structure

DEQUE: Element can be added or removed at either end but not in the middle.

AAA	BBB	CCC								
1	2	3	4	5	6	7	8	9	10	11

Priority QUEUE:

A *priority queue* is a collection of elements such that each element has been assigned a priority and such that the order in which elements are deleted and processed comes from the following rules:

- (1) An element of higher priority is processed before any element of lower priority.
- (2) Two elements with the same priority are processed according to the order in which they were added to the queue.



CSE 203: Data Structure

Data
Structure

Queue: First In First Out (FIFO)

	AAA	BBB	CCC								
	1	2	3	4	5	6	7	8	9	10	11
INSERT: DDD (where?)	AAA	BBB	CCC	DDD							
	1	2	3	4	5	6	7	8	9	10	11
DELETE: (Which one?)		BBB	CCC	DDD							
	1	2	3	4	5	6	7	8	9	10	11
DELETE: (Which one?)			CCC	DDD							
	1	2	3	4	5	6	7	8	9	10	11

It require two pointers:

FRONT to Insert

REAR to delete